

Tai Ngoc Nguyen | Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years designing and owning mission-critical distributed systems that power real-world financial outcomes currently building payment infrastructure at Intuit (TurboTax, QuickBooks) that serves millions of customers, while completing an MS in Data Science at UMass Dartmouth (May 2026). Specializes in high-throughput, low-latency backend platforms on Java 17/21, Spring Boot, and Apache Kafka; track record of shipping reliable, PII-compliant, and AI-augmented systems that scale to enterprise demand. Applies AI coding tools (GitHub Copilot, Cursor, and Claude Code) to move at startup speed without compromising quality or compliance.

TECHNICAL SKILLS

Languages: Java 17/21 (primary), Python 3.x, TypeScript/JavaScript (ES6+), SQL, Bash/Shell

Backend & APIs: Spring Boot 3.x, Spring Cloud (Config, Gateway, Circuit Breaker), Spring Security, REST API Design, GraphQL, gRPC, Microservices Architecture, API Versioning, Idempotency & Rate Limiting, Webhook Design

Cloud & Infrastructure: AWS (EKS, EC2, S3, SQS, Lambda, RDS, CloudWatch, IAM, Secrets Manager), Docker, Kubernetes, Terraform (IaC), Helm, Argo CD (GitOps), GitHub Actions, GitLab CI/CD

Databases & Messaging: PostgreSQL, MySQL, Amazon DynamoDB, MongoDB, Redis (caching & sessions), Elasticsearch, Snowflake, BigQuery | Apache Kafka (event streaming), RabbitMQ, Amazon SQS

AI/ML: PyTorch, Hugging Face Transformers, LangChain, Retrieval-Augmented Generation (RAG), GenAI Integration, OpenAI & Anthropic APIs, FAISS/Pinecone (vector search), MLflow, LLM Fine-tuning (LoRA/QLoRA), FastAPI, Flask

Observability & Security: Datadog, Prometheus, Grafana, OpenTelemetry (distributed tracing), ELK Stack, PagerDuty, SLO/SLA Monitoring | OAuth2, Keycloak, RBAC, Zero-Trust Security, OWASP Top 10, HashiCorp Vault, SOC 2, PCI-DSS Awareness, mTLS, Secrets Management

Practices: Full SDLC Ownership, SOLID Principles, Design Patterns (GoF), Resilience Engineering (Circuit Breaker, Retry, Bulkhead), Performance Tuning (JVM profiling, query optimization), TDD/BDD, Agile (Scrum/Kanban), Code Review & Technical Mentorship, RFC Authoring, On-Call & Incident Management

Architecture: Distributed Systems Design, Event-Driven Architecture, Microservices & Service Mesh, High-Availability Systems (99.99% SLA), Horizontal Auto-Scaling & Capacity Planning, Database Sharding & Replication, Fault-Tolerant System Design, Financial-Grade Transaction Processing

AI Dev Tools: GitHub Copilot, Cursor, and Claude Code integrated into daily engineering workflow for AI-assisted development, code review acceleration, and system design iteration

WORK EXPERIENCE

Software Engineer | Intuit | MA, USA

Dec 2025 – Present

- Engineered 12+ production-grade REST and GraphQL APIs in Java 17/Spring Boot, processing 5M+ financial transactions daily with full PII compliance, idempotent guarantees, and <50ms average response time.
- Diagnosed and resolved Kafka hot-partition bottlenecks in DynamoDB-backed microservices, reducing consumer lag by 40% and recovering \$200K/year in at-risk transaction throughput across distributed payment pipelines.
- Designed and implemented blue-green CI/CD pipelines (Jenkins + Argo CD) with feature-flag-controlled rollouts, increasing release cadence from bi-weekly to daily and eliminating production deployment incidents across 3 financial services teams.
- Accelerated fraud detection feature delivery by coordinating across product, design, and security teams; conducted structured root cause analyses that cut P1 incident recurrence by 30% and reduced mean time to resolution from 4+ hours to under 20 minutes.
- Authored 6+ system design documents and RFC-style architecture proposals in Agile (Scrum) sprints, establishing distributed systems standards adopted by a 15-engineer team and cutting cross-team architectural rework by 25%.

Senior Software Engineer | Vinamilk Dairy Company | Vietnam

Sep 2023 – Dec 2024

- Architected a high-availability Spring Boot microservices POS platform serving 65,000+ retail endpoints, replacing a legacy monolith to eliminate \$300K/year in vendor licensing costs, achieve 99.99% uptime SLA, and enable on-demand horizontal auto-scaling under peak load.
- Engineered a Redis caching layer and Elasticsearch search cluster that scaled product query throughput to 1M+ requests/min, slashing p99 latency from 800ms to under 100ms and supporting a 3x growth in concurrent retail users.
- Decoupled a CRM-dependent product catalog into a standalone Product Microservice with Spring Cloud Circuit Breaker, retry, and bulkhead resilience patterns, cutting inter-service error rates by 35% and improving platform-wide reliability across 600+ SKUs.
- Trained and deployed a PyTorch computer vision model for automated warehouse inventory counting across 50+ SKU categories, improving audit accuracy by 40%, eliminating 120 manual labor hours/month, and reducing reconciliation errors by 60%.

Software Engineer | CMC Global | Vietnam

Jan 2023 – Sep 2023

- Led the full SDLC migration of a monolithic IoT platform into 30+ loosely-coupled Spring Boot microservices applying SOLID principles and GoF design patterns, reducing deployment blast radius by 70% and enabling independent scaling across a 4M+ user base.
- Standardized CI/CD pipelines (GitLab, Docker, Kubernetes) with automated Datadog monitoring gates, cutting average deployment cycle time from 4 hours to under 90 minutes (45% reduction) and reducing post-release rollbacks by 60%.
- Executed a zero-downtime MySQL-to-PostgreSQL migration with schema redesign, composite index optimization, and JVM performance profiling, boosting peak-load query throughput by 30% and eliminating 3 recurring database timeout incidents per week.
- Built high-concurrency event-driven REST APIs integrating Kafka and RabbitMQ to sustain 50K+ events/sec across real-time IoT telemetry pipelines at sub-200ms p99 latency; instrumented distributed tracing with OpenTelemetry, cutting mean debug time for cross-service issues by 50%.

Java Full Stack Developer | Ban Vien Company | Vietnam

Jan 2018 – Jan 2023

- Owned end-to-end cloud migration of the CoreBar vehicle diagnostics platform from on-premise to Azure Cloud, lifting system availability from 94% to 99.9%, reducing infrastructure costs by 30% (\$80K/year), and shipping REST APIs with JavaFX interfaces covering diagnostics for 100+ car models.
- Secured enterprise platforms by implementing OAuth2/Keycloak SSO, RBAC, and zero-trust access controls aligned to OWASP Top 10, reducing unauthorized access incidents by 15%, achieving SOC 2-aligned posture, and cutting identity-related support tickets by 40%.
- Optimized 20+ high-traffic PostgreSQL queries and RESTful API endpoints using execution plan analysis and connection pooling, reducing average data retrieval time by 40% and cutting API p95 latency from 500ms to 375ms across core business services.
- Automated end-to-end Jenkins/Docker/Kubernetes CI/CD pipelines with load-balanced deployments and quality gates, cutting release cycle time by 50%, achieving 85%+ unit and integration test coverage with JUnit/Mockito, and reducing production defect escape rate by 35%.
- Mentored 3 junior engineers through weekly code reviews, pair programming, and structured PR checklists, cutting their onboarding ramp by 50% and reducing team-wide code review cycle time by 20%.
- Shipped an HR automation platform on Java Spring Boot and AWS covering onboarding, attendance, and payroll for 1,000+ employees, eliminating 15+ hours/week of manual processing, reducing admin overhead by 30%, and enforcing access control via OAuth2/Keycloak RBAC.

EDUCATION

Master's in Data Science | University of Massachusetts Dartmouth | MA, USA

May 2026

Bachelor's in Information Technology | National University HCMC — University of Science | Vietnam

Aug 2017

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate (2025)
- AWS Certified Solutions Architect – Associate (2025)

PROJECTS

Real-Time Type-Ahead Search Engine

Tech Stack: Java, Spring Boot, Python, Flask, Apache Spark, PostgreSQL, Redis

- Architected a distributed type-ahead autocomplete system with a Redis-backed trie index and consistent hashing, delivering sub-240ms query responses and supporting prefix matching across a 10M+ term dataset; full implementation available on GitHub.
- Built an Apache Spark distributed data pipeline ingesting and ranking search trend signals, refreshing top-query rankings on 24-hour rolling windows and reducing stale suggestion rates by 65%.
- Stress-tested the system under simulated 10K concurrent requests using JMeter, validated sub-240ms p99 latency at peak load, and documented architectural trade-offs across trie vs. inverted index approaches.

Real-Time Transaction Processing & Fraud Detection Platform

Tech Stack: Java, Spring Boot, Apache Kafka, Amazon DynamoDB, AWS EKS, Docker, Kubernetes

- Designed and shipped Spring Boot microservices with idempotent DynamoDB writes, eliminating duplicate transaction errors by 99% and achieving sub-50ms average processing latency at financial-grade throughput across distributed payment pipelines.
- Implemented a real-time fraud detection pipeline on Apache Kafka event streams, boosting suspicious transaction identification precision by 35%, cutting false positive rates by 28%, and diverting \$500K+/year in manual review costs to automated decisioning.
- Deployed services on AWS EKS using blue-green Kubernetes rollouts and Docker containerization, achieving zero-downtime releases across 6 production environments and cutting rollback time from 45 minutes to under 5 minutes.